

Qianwen(Kaia) Gao

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Availability: Starting May 26, 2026

SUMMARY

Data Scientist graduate student with strong foundations in applied machine learning, experimentation, and large-scale data analysis. Experienced in building and evaluating ML-driven targeting and retrieval systems using Python and modern ML frameworks. Proven ability to partner with product and business stakeholders to translate experimental insights into monetization and growth strategies.

EDUCATION

University of California, Berkeley | *Master of Computational Social Science* Berkeley, CA | Jun. 2025-Present

• **Major:** Computational Social Science | **GPA:** 4.00/4.00

Zhejiang University (ZJU) | *Bachelor of Arts*

Hangzhou, China | Sept. 2021-Jun. 2025

• **Major:** Communication | **GPA:** 3.95/4.00

SKILLS

- **Programming:** Python, SQL, R, HTML, CSS, JavaScript
- **Machine Learning:** Classification, Regression, Clustering, Predictive Modeling, Feature Engineering
- **Applied NLP:** Sentiment Analysis, Topic Modeling, BERT Embeddings, Vector Similarity, Text Classification
- **Experimentation and Statistics:** A/B Testing, Multivariate Experiments, Causal Inference, Survival Analysis
- **Frameworks and Tools:** PyTorch, scikit-learn, LangChain, W&B, Git, Jupyter Notebook, Google Colab
- **Data Analytics:** PostgreSQL, MySQL, RESTful API data ingestion, Tableau, Matplotlib, Seaborn

PROFESSIONAL EXPERIENCE

Wroodium (GEO Company) | Research Intern Berkeley, CA | Dec. 2025 - Present

- **Experimentation Framework Design** – Designed and executed a large-scale A/B/C/D experimental framework to evaluate how structured content signals such as JSON-LD and semantic HTML influence LLM-based retrieval and attribution, managing more than 5,000 longitudinal observations across 64 topics.
- **Targeting and Retrieval Evaluation** – Benchmarked retrieval and ranking performance using Recall@k, BM25 lexical ranking, and embedding-based similarity to identify content patterns that improved exposure and citation likelihood in LLM-driven search systems.
- **Applied Machine Learning Analysis** – Applied embedding models and statistical learning techniques to analyze content relevance, freshness signals, and time-to-citation dynamics in AI search pipelines.
- **Statistical Modeling for Monetization Signals** – Used Kaplan–Meier estimators and Cox Proportional Hazards models to quantify how content freshness impacts citation latency, generating insights relevant to monetization visibility and exposure timing.

Xiaohongshu | *Marketing Analytics Intern*

Shanghai, China | Aug. 2024-Jan. 2025

- **Ads Monetization Analysis** – Analyzed large-scale advertiser and user behavior data using SQL in Hive to construct 35 audience segments for pet-industry advertisers, contributing to approximately ¥1.83M (~\$250K) in incremental ad revenue.
- **KPI Automation** – Developed and automated marketing KPI dashboards using Python, SQL, and RedBI to track ad performance, user engagement, and retention metrics.
- **Targeting Performance Insights** – Combined CRM data with behavioral analysis to evaluate campaign effectiveness and targeting quality, presenting optimization insights to internal teams and more than 740 external clients.

Didi | *Product Strategy & Analytics Intern, Chauffeur Business Unit*

Hangzhou, China | Mar.-Jun. 2024

- **Pricing and Demand Modeling** – Conducted regression analysis and causal inference on supply and demand dynamics and user price elasticity, informing pricing strategies that increased revenue by 2%.
- **User Behavior Analysis** – Integrated survey findings with SQL-based behavioral data analysis to identify service pain points, reducing complaints by 3% and improving user experience.

PROJECTS

Consumer Sentiment and Brand Insights from Amazon Fashion Reviews Course Project | Oct. - Nov. 2025

- **Large-Scale Text Modeling** – Analyzed 2.5 million Amazon Fashion reviews. Engineered a hybrid pipeline using BERT embeddings for feature extraction and XGBoost regression to predict customer satisfaction ratings. Leveraged model interpretability (SHAP/Feature Importance) to identify key drivers of brand differentiation.

Consentful Civic Lens - an Event Organizer

CalHacks Project | Oct. 2025

- **Full-Stack Application Development** – Built a full-stack web application using Next.js, Supabase, and PostgreSQL, integrating LLM APIs to generate automated event summaries for downstream tasks.